ACL-903 Milestone 1

Formula 1 podium prediction

An analytical report on leakage-safe prediction after qualifying

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Team Paladin

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1. The question we are trying to answer

In this report we ask a narrow question. After qualifying, and before the race starts, can we tell which drivers will finish in the top three? We use the Formula 1 World Championship tables from 1950 to 2024. The prediction cutoff matters. Anything that is only known once the race is underway, such as the finishing order, the points scored that day, the retirement reason, the pit stops, or the fastest lap, is not allowed into the model as an input for that same race.

The cleaned table has 25,302 driver-race rows, and 13.4% of them are podiums. That rate is the reason we do not treat every metric as equal. A ranking that ignores the features would score a PR-AUC near 0.134 on the full table, and near 0.150 on the 2022-2024 seasons, because that is simply the share of positive rows. ROC-AUC answers a different question: if we pick one driver who podiumed and one who did not, how often does the model give the podium driver the higher probability? A coin-flip ranking sits near 0.500. We use PR-AUC as the main score and ROC-AUC as the secondary score. F1, precision, and recall are reported at one cutoff, and that cutoff is chosen only on 2020-2021.

> **Takeaway.** The rest of the report follows that cutoff. First we check that the tables join cleanly and that the label is one row per driver per race. Then we answer three descriptive questions about front-row starts, home races, and mechanical retirements. Only after that do we train models, and we judge them by whether they rank the rare podium rows above the rest.

2. What we checked before trusting the joins

The raw files are the 14 Kaggle/Ergast tables, and we do not edit them. The literal value \N is read as missing. Dates are parsed as dates, and qualifying times such as 1:25.187 are converted to seconds. We then audited the keys that the later joins depend on.

Table 1 covers the eleven tables with a single-column primary key. Together they hold 100,621 rows. Every audited key is non-null, and none of these keys is duplicated. That is the check we needed before treating circuitId, driverId, raceId, and the other identifiers as safe join keys.

Table 1. Primary-key audit of the raw tables. Null keys and duplicate keys are both zero in every audited table.

Table	Primary key	Rows	Null keys	Duplicate keys
circuits	circuitId	77	0	0
constructors	constructorId	212	0	0
constructor_results	constructorResultsId	12,625	0	0
constructor_standings	constructorStandingsId	13,391	0	0
drivers	driverId	861	0	0
driver_standings	driverStandingsId	34,863	0	0
races	raceId	1,125	0	0
results	resultId	26,759	0	0
qualifying	qualifyId	10,494	0	0
seasons	year	75	0	0
status	statusId	139	0	0

Table 2 checks the foreign keys we actually use. A result row should point at a real race, driver, constructor, and status. A race should point at a real circuit. Qualifying and both standings tables should point at real races. In every one of these relationships the orphan count is zero. The largest check is results.raceId, with 26,759 child rows and no missing parent.

Table 2. Foreign-key audit. An orphan is a child value with no matching parent key. None were found.

Relationship	Rows checked	Orphans
races.circuitId → circuits.circuitId	1,125	0
results.raceId → races.raceId	26,759	0
results.driverId → drivers.driverId	26,759	0
results.constructorId → constructors.constructorId	26,759	0
results.statusId → status.statusId	26,759	0
qualifying.raceId → races.raceId	10,494	0
qualifying.driverId → drivers.driverId	10,494	0
driver_standings.raceId → races.raceId	34,863	0
constructor_standings.raceId → races.raceId	13,391	0

The results file has 26,759 rows, but it is not yet one row per driver per race. Early shared drives produce more than one result

for the same driver in the same race. We found 85 such driver-race groups. The rule we keep is deterministic: prefer the row with a real grid start (grid greater than 0), then the better finishing position, then the smaller resultId. After that rule, 26,668 rows remain and each pair (raceId, driverId) appears once.

The second step removes drivers who did not enter the race. A result row with finishing code F belongs to a driver who did not qualify, did not prequalify, or failed the 107% rule. That is settled before the race starts, and the instructor ruled that these rows may be removed. We remove 1,366 of them before any feature is built, so a skipped weekend never counts as a race in a driver's form window or in the size of the field. Withdrawals (code W) stay: that code is only known after the race, and a grid of 0 can also be a real pit-lane start. 25,302 rows remain, and the seasons run from 1950 to 2024.

Grid position 0 is not a better starting spot than pole. It accounts for 6.1% of rows before the F rows are removed and 1.1% after. We keep it as a flag and treat the numeric grid position as missing in those rows, so the model cannot learn that 0 is the front of the field.

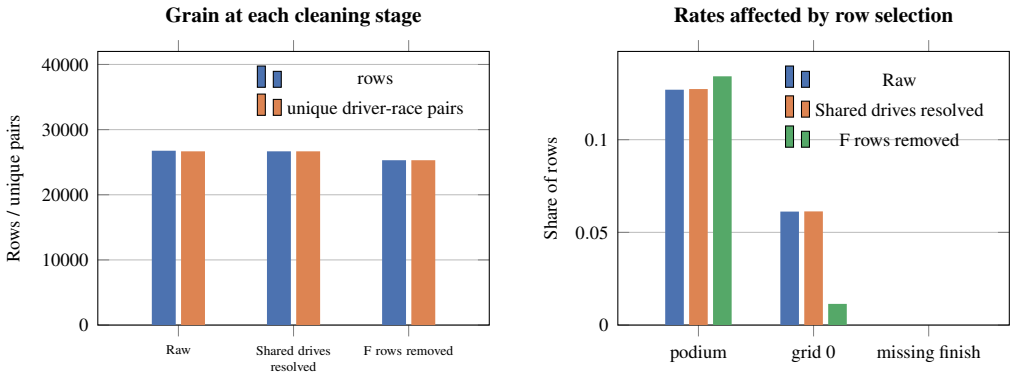


Figure 1. Programmatic EDA at each cleaning stage. The left panel compares total rows with unique driver-race pairs; the right panel checks whether each step materially changes key rates.

Table 3. Cleaning diagnostics at each stage, computed from the raw results and the final modeling table.

Stage	Rows	Unique pairs	Duplicate rows	Podiums	Grid 0	Missing finish	Median grid
Raw results	26,759	26,668	176	12.7%	6.1%	0.0%	12.0
Shared drives resolved	26,668	26,668	0	12.7%	6.1%	0.0%	12.0
Modeling table (F rows removed)	25,302	25,302	0	13.4%	1.1%	0.0%	12.0

Resolving shared drives removes 91 extra rows. Duplicate participation falls from 176 rows to zero, while the podium rate moves only from 12.69% to 12.73% and the median non-zero grid remains 12.0. So we fix the prediction grain without materially reshaping the label or the starting-position distribution. Removing the F rows takes out 1,366 more rows, and none of them is a podium, so the podium rate rises to 13.42%. That rise is arithmetic, not a change in what a podium means: a driver who never raced cannot finish in the top three. The grid-0 share falls from 6.1% to 1.1%.

> **Takeaway.** The joins we rely on later are supported by the key audit, and the modelling grain is one driver in one race who actually entered it. The cleaning choices are recorded so that a shared drive cannot silently become two training rows and a driver who never raced cannot become a row of zeros.

3. Qualifying coverage is not stable across history

Qualifying position is an obvious pre-race feature, but it is not recorded for most of the early championship. Figure 2 plots, for each season, the share of driver-race rows that have a qualifying position. We use this plot to decide whether we can drop seasons that lack qualifying, or whether we have to keep a feature that survives the whole time span.

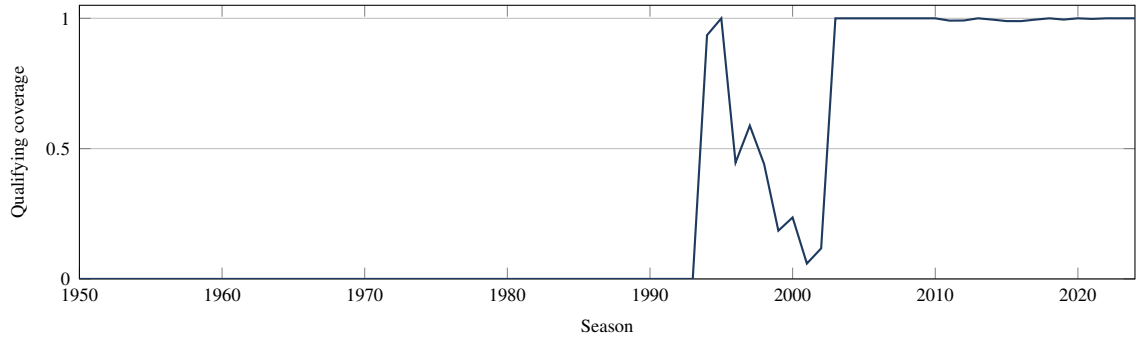


Figure 2. Share of driver-race rows with a recorded qualifying position, by season. Coverage is essentially zero until the mid-1990s, unstable until 2003, and then close to complete.

In our modelling table, coverage is 0% in 1950 and still 0% in 1993. It jumps to 93.5% in 1994 and 100% in 1995, then falls again. The low point of that second gap is 5.9% in 2001. From 2003 onwards the line sits on or next to 100%, and 2024 is 100%. Averaged over seasons, coverage is 7.2% before 2000 and 89.4% from 2000 on. At row level, 58.6% of all driver-races have no qualifying position.

We observe two different regimes, not a smooth rise. The early championship has almost no qualifying rows in this snapshot. The seasons from 1994 to 2002 are patchy. The seasons we later use for validation and test are essentially complete. Dropping the early years would throw away most of the history, and it would also make the training set look more like the test set than it really is. We keep every season, store an explicit missingness flag, and treat grid position as the starting-order feature that exists throughout. The flag lets the model see that a qualifying value was absent, rather than hiding that fact inside a filled-in number.

> **Takeaway.** Qualifying is a modern-era measurement in this dataset. Any model that depends on it alone would be fitting 2003-2024 and guessing before that. Grid position is the feature that can be compared across the whole 75 seasons.

4. Three questions about the races, before any model

The prediction model comes later. These three questions use the joined tables directly. We separate what the counts show from what we are willing to infer. None of the three answers is a causal claim.

4.1 Front-row starts convert to podiums much more often than other starts

A front-row start means grid position 1 or 2 on the final grid. We do not use qualifying position for this definition. A penalty can move a driver back between qualifying and the start, and the car that lines up first is the one we want. Results are joined to races and then to circuits. We split the history at 2014, the hybrid-era change, and we only rank a circuit inside an era when it has at least ten front-row starts there. That filter leaves 44 circuits in 1950-2013 and 19 circuits in 2014-2024.

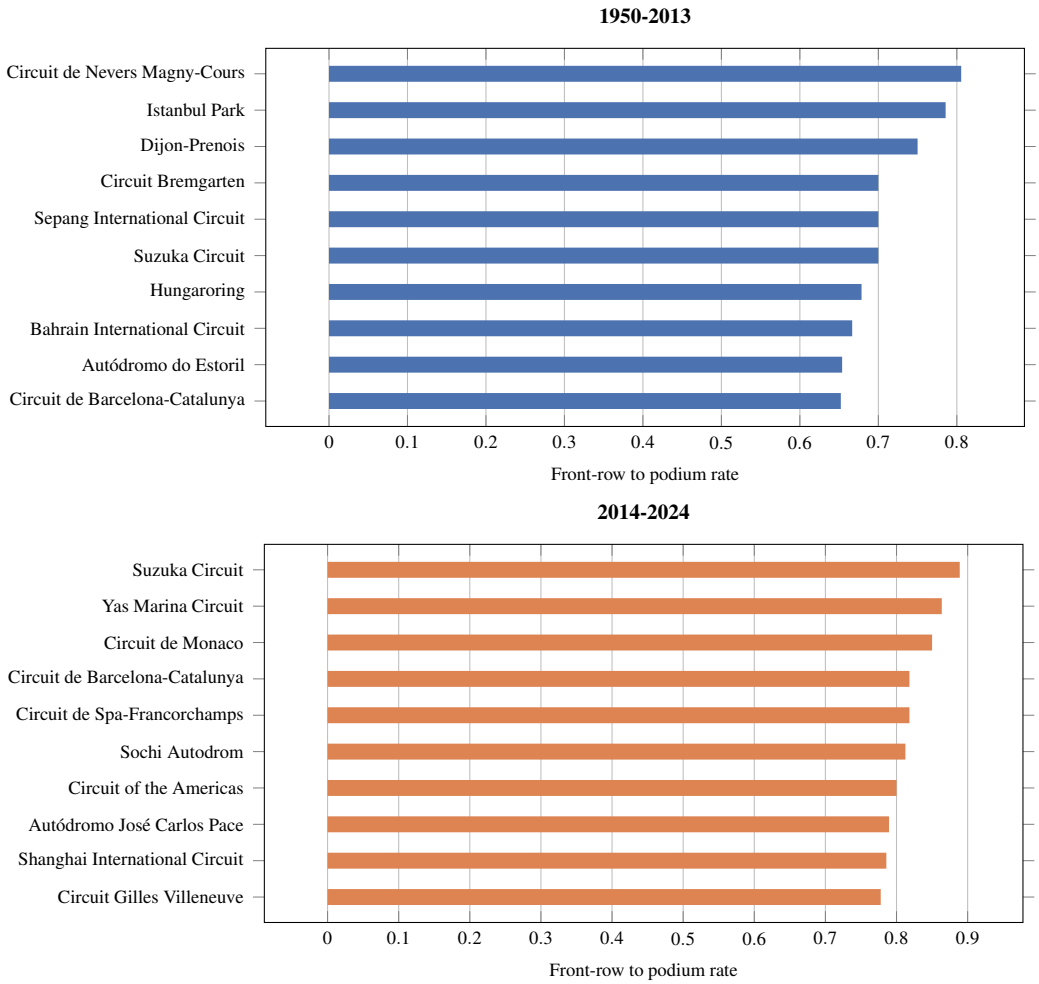


Figure 3. Ten circuits with the highest front-row podium rate in each era. The rate is podiums divided by front-row starts. Circuits with fewer than ten such starts are excluded.

Table 4. Highest front-row podium rates in each era, among circuits with at least ten front-row starts. The rate is a proportion of starts, not a mean of lap times.

Era	Circuit	Podium rate	Front-row starts
1950-2013	Circuit de Nevers Magny-Cours	80.6%	36
1950-2013	Istanbul Park	78.6%	14
1950-2013	Dijon-Prenois	75.0%	12
1950-2013	Circuit Bremgarten	70.0%	10
1950-2013	Sepang International Circuit	70.0%	30
2014-2024	Suzuka Circuit	88.9%	18
2014-2024	Yas Marina Circuit	86.4%	22
2014-2024	Circuit de Monaco	85.0%	20
2014-2024	Circuit de Barcelona-Catalunya	81.8%	22
2014-2024	Circuit de Spa-Francorchamps	81.8%	22

What we see is a shift in level, not just a change of leader. Across circuits that pass the ten-start rule, the median front-row podium rate is 56.8% in 1950-2013 and 77.8% in 2014-2024. The early range runs from 28.6% at Detroit Street Circuit (14 starts) up to 80.6% at Magny-Cours (36 starts). The modern range runs from 59.1% at Hungaroring (22 starts) up to 88.9% at Suzuka (18 starts).

If we ignore the circuit filter and use every front-row start in the modelling table, the rate is 56.0% of 1,789 starts before 2014, and 76.0% of 454 starts from 2014 on. The podium rate among all starters, not just the front row, is 13.1% and 14.8% in those same eras. A front-row car is therefore several times more likely to podium than a typical starter. The modern front row converts at a higher rate than the historical front row, while the all-starter rate only moves from 13% to 15%.

The ranking itself needs a sample-size caution. Istanbul Park is second in the early list at 78.6%, but that is 14 starts. Dijon-Prenois and Bremgarten sit near the top with 12 and 10 starts. Magny-Cours (36 starts) and Sepang (30 starts) are firmer early

observations. The next early circuit, Suzuka, is also at 70%, and it has 50 starts, so that rate is not carried only by Bremgarten. In 2014-2024 the leading circuits each have between 18 and 22 starts, so the order is less sensitive to one or two races, but Suzuka's 88.9% is still 18 starts, not a long census. We can say the leaders change with the era. We cannot say that a circuit causes the conversion rate. The front row is already occupied by the fastest cars.

> **Takeaway.** Where a driver starts dominates this descriptive cut. Era changes the level of front-row conversion, and the circuit ranking is real under our ten-start rule, with wide uncertainty on the circuits that barely meet it.

4.2 Once the grid band is fixed, home drivers do not podium more often

The second question is whether drivers do better at a home race. We map nationality to a circuit country (British to the UK, Dutch to the Netherlands, and so on) and mark a row as home when those match. Drivers whose nationality is not in the map are left out. We then compare home and away podium rates inside the same grid bands, because Section 4.1 already shows that starting position can overwhelm a smaller effect. Figure 4 plots the two series.

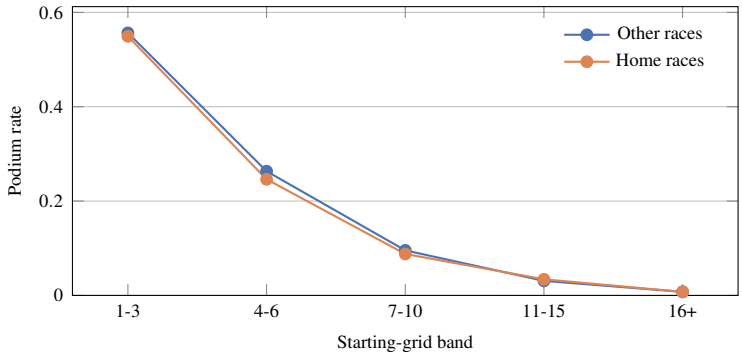


Figure 4. Podium rate by starting-grid band for home races (home_race = True) and other races (home_race = False). The two lines fall together as the start position gets worse.

Table 5. Podium rate by grid band and home status. The last column is the home rate minus the away rate, in percentage points.

Grid band	Away rate	Away starts	Home rate	Home starts	Home minus away
1-3	55.7%	3,060	54.9%	264	−0.7 pp
4-6	26.3%	3,016	24.6%	260	−1.7 pp
7-10	9.5%	3,990	8.8%	387	−0.8 pp
11-15	3.1%	4,933	3.4%	469	+0.3 pp
16+	0.8%	6,887	0.8%	933	−0.0 pp

What we find in Figure 4 and Table 5 is the grid, not the passport. Away drivers podium 55.7% of the time from positions 1-3 (3,060 starts), 26.3% from 4-6, 9.5% from 7-10, 3.1% from 11-15, and 0.8% from 16th or worse. Home drivers track those rates: 54.9%, 24.6%, 8.8%, 3.4%, and 0.8%.

The home-minus-away gaps we measure are −0.7, −1.7, −0.8, +0.3, and −0.0 percentage points. Four of the five bands are slightly negative. The only positive gap is positions 11-15, and it is +0.3 percentage points.

Home samples are also much smaller. In the front band we have 264 home starts against 3,060 away starts, so a gap under one percentage point is not a stable difference. Compared with the drop from the front band (about 55%) to the back band (under 1%), the home contrast is small.

This suggests that a raw home-versus-away average would mostly pick up who starts near the front, not a separate home advantage. The stratification is a control, not a causal model. The nationality map also misses drivers we could not assign, and it cannot represent dual nationality. We therefore report a descriptive result: inside a grid band, home drivers in this table do not show a consistent podium advantage.

> **Takeaway.** Starting position sorts podium rates into a steep ladder. Home status, measured this way, does not move a driver from one rung to another.

4.3 Mechanical retirement rates are lower in 2022-2024, and small teams sit at the top of each list

The third question is which constructors retire with a mechanical failure, and whether that rate changed between 2014-2021 and 2022-2024. We join results to status and constructor. A start counts as mechanical only when the status text is on a fixed list of component failures: engine, gearbox, brakes, hydraulics, power unit, and the other documented part failures. Crashes, driver

errors, disqualifications, and collisions are not on that list. Cars that did not start are removed from the denominator. Those statuses are Did not prequalify, Did not qualify, Did not start, Excluded, Withdrew. A constructor-era needs at least 20 starts to appear. The rate is mechanical retirements divided by starts, so a team that finishes races is in the denominator even when it rarely fails.

The included statuses are: Alternator, Battery, Brake duct, Brakes, Clutch, Cooling system, Differential, Driveshaft, Drivetrain, ERS, Electrical, Electronics, Engine, Engine fire, Engine misfire, Exhaust, Fire, Front wing, Fuel leak, Fuel pressure, Fuel pump, Fuel system, Gearbox, Hydraulics, Mechanical, Oil leak, Oil pressure, Overheating, Power Unit, Power loss, Puncture, Radiator, Rear wing, Safety belt, Spark plugs, Steering, Suspension, Technical, Throttle, Transmission, Turbo, Tyre, Tyre puncture, Undertray, Vibrations, Water leak, Water pressure, Water pump, Wheel, Wheel bearing, Wheel nut. Rare wordings that are not on this list are counted as non-mechanical. That choice can miss an ambiguous failure, and it can also avoid treating a collision as a broken part. We would rather under-count borderline cases than inflate the rate. Engine fire and engine misfire are counted as mechanical because they are engine failures; heat shield fire is not counted. None of these statuses occur in 2014-2024, so this choice does not change the rates.

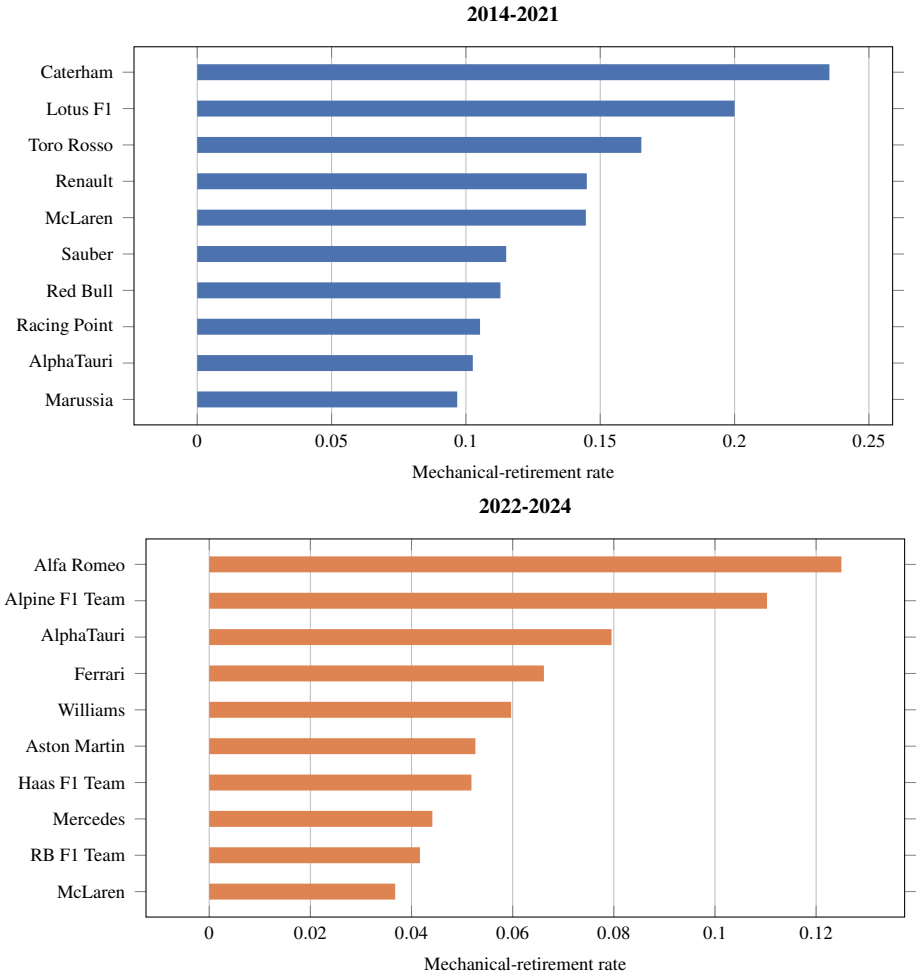


Figure 5. Ten constructors with the highest mechanical-retirement rate in each era, among those with at least 20 starts. The rate uses all starts as the denominator.

Among constructors that meet the 20-start rule, we count 329 mechanical retirements in 3,257 starts in 2014-2021 (10.1%), and 83 mechanical retirements in 1,354 starts in 2022-2024 (6.1%). The median constructor rate falls from 9.7% to 5.2%. That is 19 constructors in the first era and 12 in the second. Figure 5 shows the high end of each list. Table 6 also shows the low end, because a chart of only the worst rates makes an era look less reliable than it is.

Table 6. Highest and lowest mechanical-retirement rates in each era, plus a few large teams for scale. Rate = failures / starts, after removing non-starts.

Era	Constructor	Rate	Failures	Starts
2014-2021	Caterham	23.5%	8	34
2014-2021	Lotus F1	20.0%	15	75
2014-2021	Toro Rosso	16.5%	40	242
2014-2021	Mercedes	4.4%	14	320
2014-2021	Aston Martin	2.3%	1	44
2022-2024	Alfa Romeo	12.5%	11	88
2022-2024	Alpine F1 Team	11.0%	15	136
2022-2024	AlphaTauri	8.0%	7	88
2022-2024	Red Bull	3.7%	5	136
2022-2024	McLaren	3.7%	5	136
2022-2024	Sauber	2.1%	1	48

In our table Caterham has the highest 2014-2021 rate, 23.5%, which is 8 failures in 34 starts. Mercedes, over 320 starts, sits at 4.4% (14 failures). Aston Martin's 2.3% is 1 failure in 44 starts, so the bottom of that list is also a small sample. In 2022-2024, Alfa Romeo leads at 12.5% (11 of 88), while Sauber is at 2.1% (1 of 48). Red Bull and McLaren are both at 3.7% on 136 starts.

An interesting observation is the Alfa Romeo name across the cut: 4.2% in 2014-2021 (5 of 120 starts) against 12.5% in 2022-2024. The same constructor name does not mean the same car, and this table does not say why the rate rose. What the table does show is that the top of each era is a smaller or shorter-lived entry, while the large teams are clustered well below the leaders. The era totals are lower after 2021. One reading is the 2022 regulation change. The status strings and the 20-start filter can also move the rate, so we stop at the comparison.

> **Takeaway.** Mechanical retirements, under this definition, are less common in 2022-2024 than in 2014-2021. The identity of the highest-rate constructor changes, and those leading rates rest on fewer starts than the works teams.

5. What the model is allowed to see

The descriptive results above already point at the grid. The modelling step has to respect the cutoff. We build 16 inputs in five groups. Form uses a shift before the rolling window, so today's result is not inside today's features. Constructor form is averaged across both cars first, so one teammate cannot see the other teammate's result from the same race. Championship points and rank are taken from the previous race in the same season, because a standings row describes the table after a race. Before a driver's or team's first race of a season nothing has been scored yet, so points and rank are set to 0 there. Form before a driver's first race has no history, and we leave it missing rather than filling it with the future.

Table 7. Feature groups. Post-race position, points scored today, laps, status, pit stops, and fastest lap are excluded.

Group	Inputs	Why it is known before the start
Grid	Grid position, pit-lane flag, front-row flag, field size	Final grid is set before lights out
Qualifying	Position, best Q1/Q2/Q3 time, missingness flag	Qualifying is already finished
Form	Driver and constructor podium rate over the previous 5 races, prior starts	Current race is shifted out
Standings	Driver and constructor points and rank	Attached from the previous race only
Context	Driver age, year	Known on race morning

We split by season, not at random. A random split would let 2023 teach the model about 2022. Training is every season through 2019, validation is 2020-2021, and the test seasons 2022-2024 are scored once. Training fits the imputer, scaler, and weights; validation makes every choice (preprocessing, ablation, model, cutoff); test is only reported. Table 8 shows why that split is not a copy of the same problem three times.

Table 8. Time split used for every model. Starters per race is the number of modelling rows divided by races.

Split	Seasons	Rows	Races	Starters / race	Podium rate	Qualifying missing
Training	1950-2019	23,163	1,018	22.8	13.3%	64.0%
Validation	2020-2021	780	39	20.0	15.0%	0.1%
Test	2022-2024	1,359	68	20.0	15.0%	0.0%

Our training seasons have 23,163 rows and 1,018 races, but 64.0% of those rows have no qualifying result. Validation has only 780 rows and 39 races, and the test set has 1,359 rows and 68 races. Qualifying is almost complete in both recent splits (0.1%

missing in validation, 0% in test). The podium rate rises from 13.3% in training to 15.0% and 15.0% later. That matches the field size: training averages 22.8 starters per race, against 20.0 in validation. There are still three podium places, shared across fewer cars.

All 7 of our split checks pass: every row is assigned once, no race appears in two splits, and the last training race is before the first validation race, which is before the first test race. Figure 6 shows the split on the season axis together with the two quantities that drift across it.

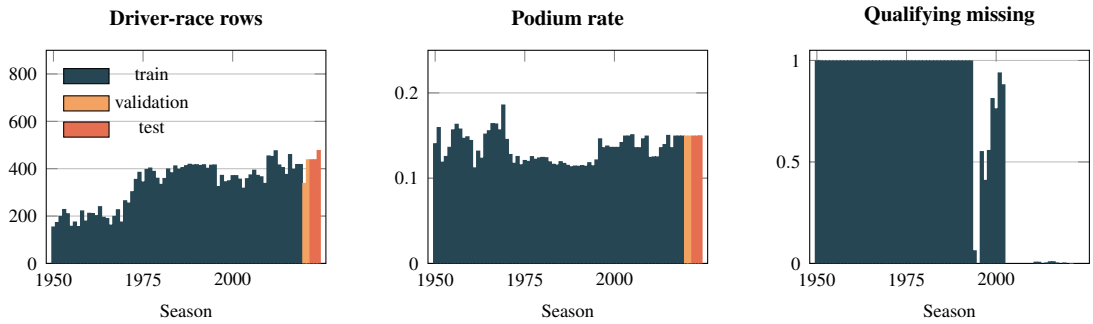


Figure 6. Rows, podium rate, and qualifying missingness by season, coloured by split.

To check our choice, Table 9 holds out the same number of rows three ways, and we train the logistic baseline on the rest. With a random-row split, 100.0% of held-out rows have their own race in training, and 98.2% have later seasons in training. A random-race split removes the first problem but not the second. The chronological split has zero of both. Random hold-outs also mix every era, so their score (0.588 PR-AUC for random rows, against 0.730 on 2020-2021) answers a different question from predicting the next seasons.

Table 9. Split-strategy check with logistic regression. Leakage columns are shares of held-out rows.

Strategy	Held-out years	Podium rate	Same race in train	Later seasons in train	PR-AUC	ROC-AUC
chronological (used)	2020-2021	15.0%	0.0%	0.0%	0.730	0.926
random races	1952-2023	13.3%	0.0%	100.0%	0.627	0.920
random rows	1950-2024	11.0%	100.0%	98.2%	0.588	0.901

From these results, we notice that validation is a short window. Two seasons and 39 races can separate a large gap from a small one, but they cannot support a fine ranking of models that differ by a few thousandths. We still use it for the decisions the test set must not make: the probability cutoff, and which model is used in the examples.

> **Takeaway.** The features are pre-race by construction, and the split is chronological. The training years are a different problem from 2020-2024: more missing qualifying, larger fields, and a lower podium rate.

5.1 Sprint races: known before the start, but not a feature

A sprint is a separate short race on the same weekend, stored in the sprint-results table. We had to decide whether it leaks. By time it does not: every sprint ends one day before the main race, so its result is known before the lights go out. We still leave it out, for two reasons. Sprints exist only from 2021, so no training row (1950-2019) has one; there are 0 such rows. And in 2021-2022 the sprint set the main-race grid, so the grid features already contain it (Table 10).

Table 10. Sprint weekends. The last two columns are Spearman rank correlations of the main-race grid with the sprint finishing order and with qualifying position.

Season	Sprint races	Driver rows	Split	Days before race	Grid vs sprint finish	Grid vs qualifying
2021	3	60	validation	1	0.89	0.83
2022	3	60	test	1	0.99	0.72
2023	6	120	test	1	0.62	0.99
2024	6	120	test	1	0.54	0.96

In 2023-2024 the sprint stopped setting the grid, which follows qualifying again. A sprint feature would then cover only 12 weekends, so we leave it out. The hazard we guard against is a join: sprint positions and points sit under the same race identifier as the main race, but they belong to a different race. Standings are already moved to the following round, so sprint points reach the features only through the previous round.

5.2 Drivers who did not enter the race

Before this step 6.1% of rows have grid 0, and the feature name suggests a pit-lane start. Most of them were not. 1,366 rows are drivers who did not qualify, did not prequalify, or failed the 107% rule. All of them are in 1952-2012, all are in training, and none finished on the podium. Only 87 grid-0 rows are real pit-lane or exceptional starts (Table 11). The instructor ruled that these F rows are known before the race and may be removed, and that withdrawals may not, so we remove the first and keep the second.

Table 11. Who has grid 0 or a withdrawal code, by group, and what we do with each group.

Group	Treatment	Rows	Podium rate	Years	In training	In validation	In test
regular grid start	kept	24,880	13.6%	1950-2024	22,806	759	1,315
grid 0 but raced (pit-lane or exceptional start)	kept	87	0.0%	1981-2024	34	16	37
W: withdrew	kept	335	0.0%	1950-2024	323	5	7
F: did not qualify, prequalify, or pass 107%	removed	1,366	0.0%	1952-2012	1,366	0	0

We scored the decision on the same validation seasons. Keeping the F rows instead changes validation PR-AUC by -0.0005 for logistic regression and +0.0144 for the network. Keeping them in the history but dropping them from the training rows only changes it by -0.0001 and +0.0126. Logistic regression, which has no random seed, moves by at most 0.0005. The network moves by up to 0.0144, but one network seed can sit 0.021 away from its own three-seed average (Section 5.4), so the network cannot separate these options. The choice rests on the ruling and on what a row means, not on a score. Removing the withdrawals as well, which the ruling does not allow, would change the network by +0.0182. Validation and test contain no F rows, so the same rows are scored either way. The pit-lane flag now marks 289 rows, 202 withdrawals with grid 0 and 87 drivers who raced from the pit lane or an exceptional start. It is no longer a did-not-qualify marker, but it is still not a pure pit-lane flag.

5.3 Evidence for each individual feature

The group-level ablation in Section 8 says what a whole group adds to the model. Each feature also has its own exploratory evidence, computed on the training seasons only (Table 12 and Figure 7). For every feature we report the share of missing cells, the podium rate in its lowest and highest bin, and a one-feature ROC-AUC. An AUC above 0.5 means a higher value goes with more podiums, and below 0.5 the reverse. The same AUC on validation checks that the pattern still holds in later seasons.

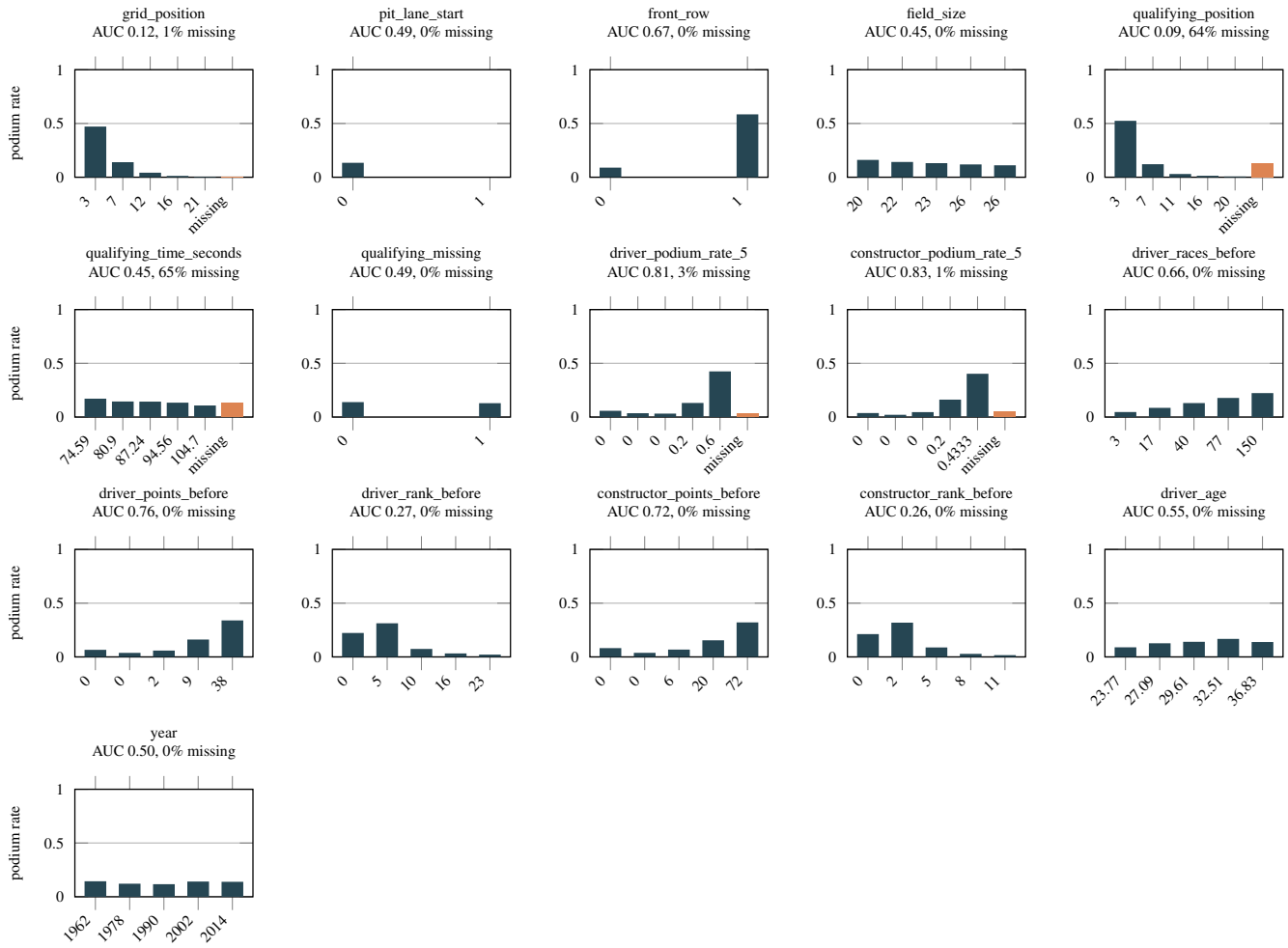


Figure 7. Podium rate by feature value on the training seasons. Each bar is one of five equal-count bins (labelled by its median value; tied values are split across bins, so a label can repeat), flags show 0 and 1, and the orange bar is the podium rate of the rows where the feature is missing.

Table 12. One-feature evidence on the training seasons. Missing cells are left out of the bins and the AUC. Valid. is the same AUC on 2020-2021.

Feature	Group	Missing	Lowest bin	Highest bin	AUC train	AUC valid.
grid_position	grid	1%	47.1%	0.5%	0.12	0.09
pit_lane_start	grid	0%	13.4%	0.0%	0.49	0.49
front_row	grid	0%	8.9%	58.5%	0.67	0.73
field_size	grid	0%	16.1%	11.1%	0.45	0.50
qualifying_position	qualifying	64%	52.5%	0.7%	0.09	0.08
qualifying_time_seconds	qualifying	65%	17.1%	10.8%	0.45	0.46
qualifying_missing	qualifying	0%	13.9%	12.9%	0.49	0.50
driver_podium_rate_5	form	3%	5.8%	42.5%	0.81	0.84
constructor_podium_rate_5	form	1%	3.7%	40.3%	0.83	0.85
driver_races_before	form	0%	4.7%	22.3%	0.66	0.70
driver_points_before	standings	0%	6.6%	34.0%	0.76	0.80
driver_rank_before	standings	0%	22.3%	2.1%	0.27	0.15
constructor_points_before	standings	0%	8.2%	32.1%	0.72	0.79
constructor_rank_before	standings	0%	21.2%	1.6%	0.26	0.16
driver_age	context	0%	8.9%	13.9%	0.55	0.60
year	context	0%	14.4%	13.9%	0.50	0.50

Strong single signals. Position features carry the most. Qualifying position has an AUC of 0.09 (lower is better, measured on the 36% of rows that have it) and grid position 0.12; the podium rate falls from 47% (grid) and 52% (qualifying) in the best bin to 0.5% and 0.7% in the worst. The standings ranks (0.27 and 0.26), the form rates (0.81 and 0.83), and the points (0.76 and 0.72) come next. The front-row flag (0.67) and prior starts (0.66) are clear but smaller. All of these keep their direction on validation, with similar or slightly stronger AUCs.

Weak single signals, kept for another reason. Field size (0.45), driver age (0.55), qualifying time (0.45), the pit-lane flag

(0.49), and the qualifying-missing flag (0.49) are close to 0.5 on their own. A raw qualifying time is not comparable across circuits and eras, so it says little without the rest of the row. The two flags are indicators. Removing the qualifying-missing flag changes PR-AUC only slightly (Section 6.4), but the flag records which rows were imputed, and rows with no grid position (withdrawals and pit-lane starts) never podium, while rows with no qualifying record podium at about the average rate (12.9% against 13.4%). Year (0.50) has essentially no single-feature signal, and field size and year are flat or constant on validation, where every race has 20 starters. The ablation shows the whole context group adds the least of the five (0.001 PR-AUC). Section 5.4 tests the two calendar features one at a time: year is kept and round was removed.

> **What this does and does not show.** A one-feature AUC ignores the other 15 features. A feature that looks weak here can still help in combination, and a strong one can add little once the grid is present: the ablation shows qualifying adds only 0.006 on top of the grid. The AUCs describe association in past races, not causes.

5.4 Calendar features: keep year, remove round

Year and round were added as context: the era and the point in the season. On their own neither separates podiums from the rest (year has a single-feature AUC of 0.50), so we tested each one inside the full models. Every variant is trained on 1950-2019 and scored on 2020-2021 only, and the network is averaged over three seeds (Table 13).

Table 13. Calendar-feature experiment on 2020-2021. Change is against the features we use (no round).

Variant	Model	Inputs	Val. PR-AUC	Seed sd	Seed 42	Change
used (no round)	Logistic regression	16	0.7301	0.0000	0.7301	+0.0000
used (no round)	Random forest	16	0.7125	0.0000	0.7125	+0.0000
used (no round)	Shallow neural network	16	0.7228	0.0052	0.7166	+0.0000
put round back	Logistic regression	17	0.7293	0.0000	0.7293	−0.0008
put round back	Random forest	17	0.7161	0.0000	0.7161	+0.0037
put round back	Shallow neural network	17	0.7232	0.0170	0.7443	+0.0003
also remove year	Logistic regression	15	0.7307	0.0000	0.7307	+0.0006
also remove year	Random forest	15	0.7080	0.0000	0.7080	−0.0045
also remove year	Shallow neural network	15	0.7017	0.0157	0.7108	−0.0212
round back, year removed	Logistic regression	16	0.7289	0.0000	0.7289	−0.0012
round back, year removed	Random forest	16	0.7080	0.0000	0.7080	−0.0045
round back, year removed	Shallow neural network	16	0.6869	0.0446	0.7167	−0.0360

Round adds nothing, so we removed it. Putting round back changes validation PR-AUC by −0.0008 for logistic regression, +0.0037 for the forest and +0.0003 for the network. All three are far smaller than the network's seed spread of up to 0.017, and they do not agree in direction. The standings already carry the same information, because points and rank build up race by race. Round also changes meaning across eras: a season had about 8 races in the 1950s and 24 in 2024, so round 8 was a finale then and an early race now. With no measurable gain and an unstable meaning, the simpler model wins.

Year helps, so we kept it. Removing year changes the score by +0.0006 for logistic regression, −0.0045 for the forest and −0.0212 for the network. The network loses the most of the three models, about 2 times the noise of a three-seed average (about 0.010). The same test inside the old 17-feature model, with round still there, gives −0.0004, −0.0082 and −0.0363, so year helps the nonlinear models with or without round. That old model without year is also unstable: its network seed spread is 0.045. Year alone says nothing, but together with grid and form it lets the nonlinear models adjust for the era: larger fields, worse reliability, and missing qualifying in older seasons. One caveat: the test seasons lie outside the training years, so the models must extrapolate year, and a tree cannot, so the forest treats 2022 like 2019.

> **Takeaway.** One seed is not enough to rank the models. With round, the network's seed-42 score was 0.7443, 0.021 above its own three-seed average of 0.7232. Without round, seed 42 scores 0.7166 and the average is 0.7228. The average barely moved, yet the single seed fell, and that alone moved the network from first to below logistic regression in Section 7. We therefore read single-seed network differences of about 0.021 as noise throughout this report.

6. Feature engineering and preprocessing: strategy, effect on the values, and order

Missing qualifying, grid, and form values have to be filled inside the model pipeline, using training rows only. We compare three settings of logistic regression on the validation seasons: median imputation with scaling, median imputation without scaling, and filling missing values with zero before scaling. The test seasons are not used for this choice.

Table 14. Preprocessing comparison for logistic regression on 2020-2021 only.

Preprocessing	Validation ROC-AUC	Validation PR-AUC
median + scaling	0.926	0.730
median, no scaling	0.926	0.731
zero + scaling	0.927	0.726

We can see that the three PR-AUC values sit between 0.726 and 0.731. The gap is 0.005. Median imputation with scaling is the best of the three on the metric we care about. Zero-filling with scaling is slightly higher on ROC-AUC (0.927 against 0.926). If we had followed ROC-AUC, we would have picked a different setting. The difference is small enough that we do not treat preprocessing as the main lever. We keep median imputation with scaling for the linear model and the network. The forest does not use scaling, because a tree split does not depend on the units of a feature.

6.1 What feature engineering does to the values

In Table 15 we compare each raw source column with the feature we built from it. Raw grid has a minimum of 0 and 1.1% zeros, which would rank a pit-lane start ahead of pole. In the grid-position feature those zeros become missing, so the minimum is 1, and the pit-lane flag carries the information instead. Qualifying strings become seconds; after the join to every driver-race row, 59.2% of rows have no qualifying time. Shifted form is missing on a driver's first race, because nothing earlier exists, and that gap is what the imputer fills next. Previous-race standings are not imputed. Before a driver's or team's first race of a season they used to be empty, and the median imputer filled them with about 3 points and rank 13, points nobody had scored. They are now set to 0 points and rank 0. Extra flags marking these rows gave no validation gain, so no flags were added.

Table 15. Raw source columns against the engineered features built from them (all 1950-2024 rows).

Column	Missing	Zeros	Min	Median	Max
grid (raw)	0.0%	1.1%	0.00	11.00	34.00
grid_position (0 → missing)	1.1%	0.0%	1.00	12.00	34.00
pit_lane_start (grid == 0)	0.0%	98.9%	0.00	0.00	1.00
q1/q2/q3 text present	0.0%	1.5%	0.00	1.00	1.00
best qualifying seconds (parsed)	1.5%	0.0%	53.38	86.69	1002.64
qualifying_time_seconds (joined)	59.2%	0.0%	53.38	86.69	1002.64
driver_podium_rate_5 (shifted)	3.2%	64.7%	0.00	0.00	1.00
driver_points_before (previous race)	0.0%	40.8%	0.00	3.00	549.00

6.2 What imputation and scaling do to the values

In Table 16 we measure both steps on the training rows. Median imputation writes the training median into every gap. It does not change the range, because the median is inside it, but it piles rows onto one value and shrinks the spread. qualifying_time_seconds is missing in 64.5% of training rows, and its standard deviation falls by 40.3%. The qualifying-missing flag keeps the fact that a value was filled. Scaling then puts every training feature at mean 0 and standard deviation 1. Raw ranges run from 0-1 for the flags to 722 constructor points; after scaling they share one scale, so neither the logistic penalty nor the network's gradient steps are dominated by large units. Scaling is linear, so the shape and order of values do not change.

Table 16. Effect of median imputation and scaling, fitted on 1950-2019. Val. and test means are after the same training-fitted transform; training means are 0 by construction.

Feature	Missing	Fill value	Raw range	Std change	Scaled range	Val. mean	Test mean
grid_position	1.0%	12.00	1.0 to 34.0	−0.5%	−1.60 to 3.21	−0.24	−0.25
qualifying_position	64.0%	11.00	1.0 to 26.0	−39.9%	−2.65 to 3.89	−0.17	−0.17
qualifying_time_seconds	64.5%	87.24	63.0 to 1002.6	−40.3%	−2.63 to 97.17	−0.45	−0.25
driver_podium_rate_5	3.5%	0.00	0.0 to 1.0	−1.2%	−0.57 to 3.77	+0.08	+0.08
driver_points_before	0.0%	2.00	0.0 to 387.0	0.0%	−0.42 to 10.85	+0.95	+1.30
constructor_points_before	0.0%	6.00	0.0 to 722.0	0.0%	−0.44 to 10.43	+0.99	+1.34
driver_age	0.0%	29.61	17.5 to 56.8	0.0%	−2.42 to 5.09	−0.43	−0.25
year	0.0%	1990.00	1950.0 to 2019.0	0.0%	−2.09 to 1.59	+1.67	+1.80

Figure 8 shows the spike at the median and the shift of later seasons; in Figure 9 we compare raw and scaled ranges for all 16 features. Because the transform is fitted on training only, validation and test are not centred at zero: year sits at +1.67 standard deviations in validation. That is the period shift the model faces at deployment, not a bug.

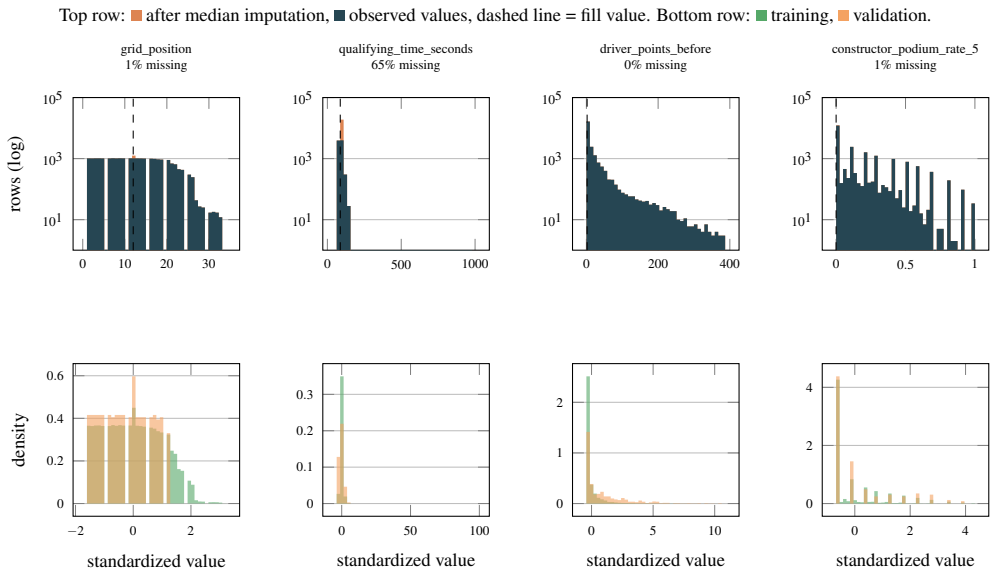


Figure 8. Top: training values before and after median imputation (dashed line is the fill value, log scale). Bottom: training and validation after the training-fitted scaler.

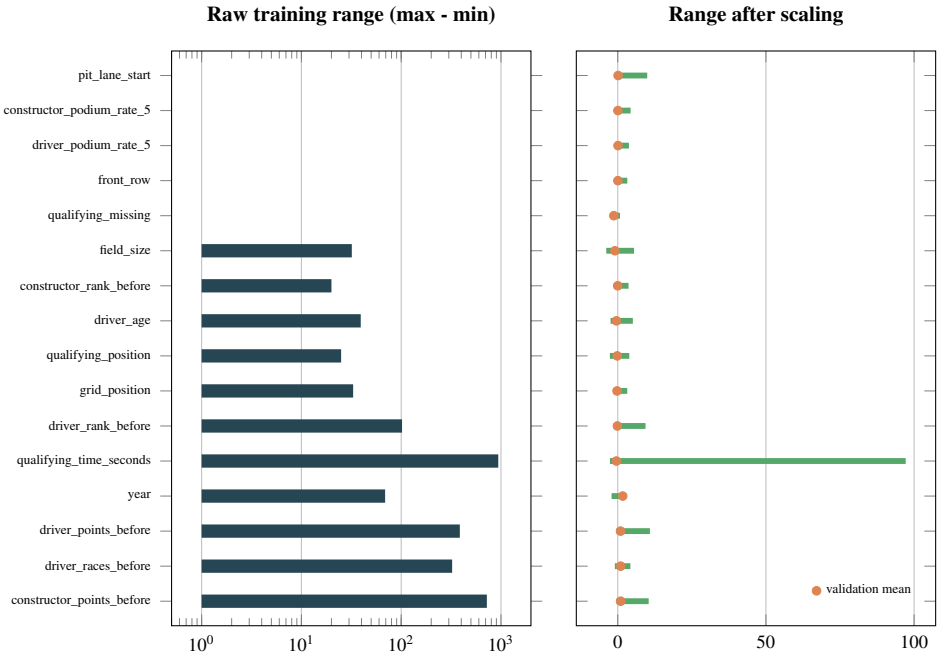


Figure 9. Left: raw training range per feature. Right: range after scaling, with the validation mean.

6.3 Preprocessing-order experiments

The table above changes which strategy we use. We also ran three experiments that change the *order* of the steps (Table 17), each scored on validation with logistic regression and the network. A swaps median imputation and scaling. B swaps zero-filling and scaling. C fits the preprocessing on all seasons before splitting, against the normal fit on training only.

Table 17. Preprocessing-order experiments on 2020-2021. Change is against the first order of the same experiment and model; for C it is against the train-only fit.

Experiment	Order	Fitted on	Model	Missing grid becomes	Val. PR-AUC	Change
A	impute → scale	train	LogReg	12.00	0.730	+0.0000
A	impute → scale	train	MLP	12.00	0.717	+0.0000
A	scale → impute	train	LogReg	12.00	0.730	+0.0000
A	scale → impute	train	MLP	12.00	0.733	+0.0164
B	zero-fill → scale	train	LogReg	0.00	0.726	+0.0000
B	zero-fill → scale	train	MLP	0.00	0.740	+0.0000
B	scale → zero-fill	train	LogReg	12.00	0.730	+0.0043
B	scale → zero-fill	train	MLP	12.00	0.735	−0.0054
C	impute → scale	all years	LogReg	12.00	0.730	+0.0001
C	impute → scale	all years	MLP	12.00	0.737	+0.0208

A. Median imputation commutes with a linear rescale, so a missing grid becomes 12.0 either way. Only the scaler's statistics differ, because scaling first measures spread on observed values. Validation PR-AUC moves by +0.0000 for logistic regression and +0.0164 for the network. **B.** Here order changes the value. Zero-filling first writes a literal 0, a grid slot ahead of pole and a 0-second lap. Zero-filling after scaling writes the training mean (12.0 for grid). Validation PR-AUC changes by +0.0043 (logistic) and −0.0054 (network). **C.** Fitting before the split lets 2020-2024 set the medians and scaling used in training. The score moves by +0.0001 and +0.0208; small or not, it uses future data, so we never do it.

> **Takeaway.** We keep split, then median imputation, then scaling, all fitted on training rows inside the model pipeline. Logistic regression is almost indifferent to strategy and order (every change is within 0.005 PR-AUC). The network moves by up to 0.021, about the 0.021 that one seed can move on its own (Section 5.4), so its numbers cannot rank the orders either. We fix the order on principle: fit on training rows only, and impute before scaling so the scaler sees the filled values. The next sections test which information the models need.

6.4 Missing values: impute, drop columns, or drop rows?

About 64% of training rows have no qualifying record, so we did not choose the handling by preference. Each approach is scored on the same validation seasons (Table 18). The network is run once with one seed, and one seed can sit 0.021 away from its own average (Section 5.4), so network differences of that size are noise. We decide on logistic regression, the selected model, because it has no random seed.

Table 18. Missing-value experiments on 2020-2021. Change is against median imputation, which is used.

Approach	Training rows	LogReg PR-AUC	Change	MLP PR-AUC	Change
median imputation (used)	23,163	0.7301	+0.0000	0.7166	+0.0000
mean imputation	23,163	0.7304	+0.0003	0.7066	−0.0100
KNN imputation (k=5)	23,163	0.7277	−0.0024	0.7326	+0.0160
iterative (regression) imputation	23,163	0.7280	−0.0021	0.7336	+0.0169
median, without the qualifying_missing flag	23,163	0.7294	−0.0007	0.7113	−0.0053
drop the sparse columns (qualifying_position, qualifying_time_seconds)	23,163	0.7242	−0.0059	0.6861	−0.0305
drop training rows with no qualifying (keeps 1994+)	8,348	0.7290	−0.0011	0.7055	−0.0111
drop training rows with any missing feature	8,054	0.7314	+0.0014	0.7387	+0.0221

For logistic regression every alternative stays within 0.006 of median imputation (the best alternative, drop training rows with any missing feature, is +0.0014). For the network, mean imputation changes the score by −0.010, k-nearest-neighbour imputation by +0.016, and iterative imputation by +0.017. Dropping the two sparse qualifying columns changes the network by −0.030, the one network change clearly beyond single-seed noise, and removing the missing-value flag instead changes it by −0.005. Dropping rows keeps only 8,348 or 8,054 of 23,163 training rows and changes the scores by −0.001 and +0.001 for logistic regression, −0.011 and +0.022 for the network. **We keep median imputation with the flag**, fitted on training rows only: for logistic regression it is tied with every alternative, the network's single-seed numbers cannot rank them, and it keeps all the history.

7. Three models, and why the scores are not interchangeable

We fit three classifiers on the same training rows and the same 16 features. Logistic regression is the linear baseline, with class weights so the rare podium rows are not ignored. The random forest has 240 trees, a depth limit of 12, and a minimum of 30

rows per leaf; it can use combinations such as a strong recent record together with a poor grid. The neural network has two hidden layers, of 32 and 12 units. Each model's settings were searched on validation (Section 7.1), and the probability cutoff that maximises F1 on validation is chosen last. That produces a different cutoff for each model, because the three probability scales are not the same number line. A cutoff of 0.455 on the network is not automatically more aggressive than 0.625 on the logistic model. Precision and recall tell us what the cutoff actually does.

Figure 10 compares the three models on the test seasons only. Table 19 gives training, validation, and test, so we can see whether a high training score survives.

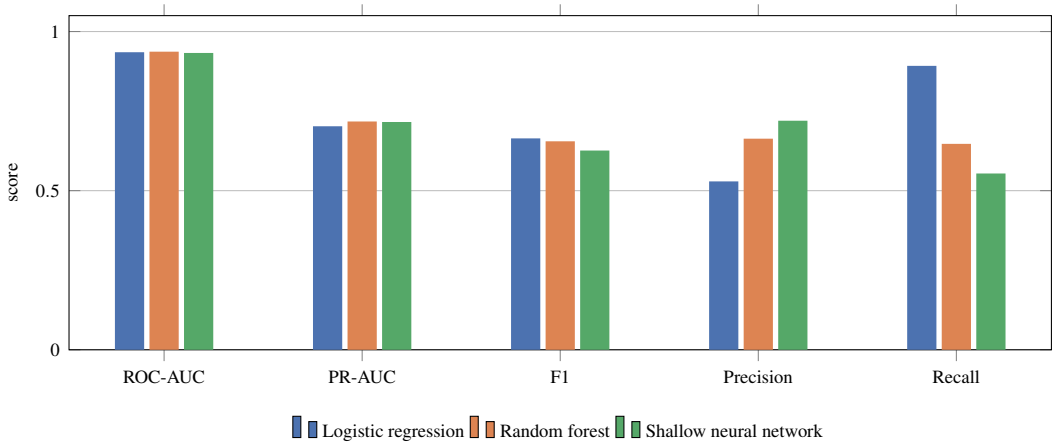


Figure 10. Test-set scores for 2022-2024. ROC-AUC is similar across models. Precision and recall differ because each model uses its own validation cutoff.

Table 19. Scores by model and split. The cutoff maximises F1 on validation and is then frozen for training and test.

Model	Split	Cutoff	ROC-AUC	PR-AUC	F1	Precision	Recall
Logistic regression	train	0.625	0.899	0.562	0.578	0.461	0.773
Logistic regression	validation	0.625	0.926	0.730	0.700	0.630	0.786
Logistic regression	test	0.625	0.935	0.702	0.664	0.529	0.892
Random forest	train	0.775	0.927	0.646	0.604	0.607	0.600
Random forest	validation	0.775	0.925	0.712	0.701	0.711	0.692
Random forest	test	0.775	0.937	0.717	0.655	0.663	0.647
Shallow neural network	train	0.455	0.902	0.576	0.525	0.603	0.465
Shallow neural network	validation	0.455	0.922	0.717	0.710	0.784	0.650
Shallow neural network	test	0.455	0.933	0.716	0.626	0.720	0.554

On the test set we measure a ROC-AUC of 0.935 for logistic regression, 0.937 for the forest, and 0.933 for the network. Those three numbers are within 0.005 of each other. In the pair reading of ROC-AUC, a true podium finisher outranks a non-podium about 93 to 94 times out of 100. That is a strong ranking. It is also a ranking on which the models do not separate. ROC-AUC compresses the comparison because most rows are not podiums, so a modest number of false alarms barely moves the false-positive rate.

PR-AUC is where they separate a little. On test, the forest scores 0.717, the network 0.716, and logistic regression 0.702. The test podium rate is 15.0%, so a useless ranking would sit near 0.150. All three models are far above that floor. The forest's lead over logistic regression is 0.015 PR-AUC. We call that a small lead, not a different kind of model. We selected logistic regression before looking at this comparison, because its validation PR-AUC of 0.730 was ahead of 0.717 for the network and 0.712 for the forest. It also leads on the three-seed average (Section 7.1), so the choice does not rest on one seed. The test seasons are never used to choose.

The operating points we chose explain the precision and recall bars in Figure 10. At its validation cutoff, logistic regression has test precision 0.529 and recall 0.892. It catches most podiums and accepts more false alarms. The forest has precision 0.663 and recall 0.647, and the network has precision 0.720 and recall 0.554. The network is the conservative one of the three. Test F1 ends up almost tied, from 0.626 to 0.664. Same F1, different mix of missed podiums and false podiums.

The training column is the check for memorizing. All three models have a training PR-AUC below their validation score: 0.562 for logistic regression, 0.646 for the forest, and 0.576 for the network. That was not true of the first forest, which scored 0.917 on training and 0.709 on validation because it had no depth limit and four rows per leaf. The validation search in Section 7.1 limited it. We do not read the higher validation score as the model improving after 2019. Training has a lower podium rate and far more missing qualifying, so the PR-AUC floor on training is lower as well. The comparison we trust is validation against test, where the positive rate is almost the same.

> **Takeaway.** All three models rank podium finishers well above non-podiums. On the primary score, logistic regression is slightly ahead on validation and slightly behind on test, and both gaps are smaller than the uncertainty of the race samples (Section 7.2). ROC-AUC hides even that difference.

7.1 Searching each model's settings, on validation only

We trained each model on 1950-2019 and scored it on 2020-2021 for every setting in a small grid. Our grid held 8 settings for logistic regression, 9 for the forest, and 12 for the network. We averaged the network over three seeds because its score depends on the initialization. We did not use the test seasons (Table 20).

Table 20. Search results for the starting settings, the adopted settings, and the top of each grid.

Model	Setting	Role	Val. PR-AUC	Seed sd	Train PR-AUC
Logistic regression	C: 1.0, penalty: l1	top of grid	0.7303	0.0000	0.5623
Logistic regression	C: 1.0, penalty: l2	starting, kept	0.7301	0.0000	0.5624
Random forest	max_depth: None, min_samples_leaf: 12	top of grid	0.7160	0.0000	0.7369
Random forest	max_depth: 12, min_samples_leaf: 30	adopted	0.7125	0.0000	0.6456
Random forest	max_depth: None, min_samples_leaf: 4	starting	0.7088	0.0000	0.9168
Shallow neural network	hidden_layer_sizes: (64,), alpha: 0.002	top of grid	0.7362	0.0121	0.5869
Shallow neural network	hidden_layer_sizes: (32, 12), alpha: 0.02	adopted	0.7228	0.0052	0.5813
Shallow neural network	hidden_layer_sizes: (32, 12), alpha: 0.002	starting	0.7179	0.0051	0.5769

Random forest: improved. The starting forest (no depth limit, 4 rows per leaf) ranked 5th of 9. It scored 0.709 on validation but 0.917 on training, a memorized fit. A depth limit of 12 with 30 rows per leaf scores 0.712 and ranks 2nd of 9. The best setting (max_depth: None, min_samples_leaf: 12) is only 0.004 higher, far inside the validation noise in Section 7.2, so we keep the depth limit as the more constrained model. It cuts the training score to 0.646, so it no longer memorizes the training seasons. No test season took part in the choice. **Shallow network: a noisy search.** One layer of 8, 16, 32, or 64 units and two layers of 32 and 12 or 64 and 32, each with two regularization strengths, were all tried and averaged over three seeds. The adopted two layers of 32 and 12 with alpha 0.02 score 0.7228 and rank 7th of 12, against 0.7179 for the starting alpha of 0.002. The best setting (hidden_layer_sizes: (64,), alpha: 0.002) is 0.013 higher, while one setting's seed spread reaches 0.025 in this search. The ranking is also unstable: with round in the features, the adopted setting ranked first. Picking the top of a ranking that moves with one input and one seed would be picking noise, so we keep the adopted network. Networks with 8 units are clearly worse, so it is not a deliberately weak choice. Its three-seed average is still below logistic regression's 0.7301. **Logistic regression** is insensitive: L1 and L2 penalties and C from 0.01 to 10 all fall between 0.710 and 0.730, and the starting C = 1 with L2 ties for best (within 0.0005), so we kept it.

7.2 How much the validation score can be trusted

We also asked whether 2019 is a good boundary. The same two models were scored on four validation windows that all end before the test seasons, each training on every earlier season (Table 21). The score depends on the window: logistic regression ranges from 0.689 to 0.730, and the network from 0.714 to 0.740. The 2020-2021 window we use ranks 1st of 4 for logistic regression and 3rd of 4 for the network, so it sits near the top of both ranges and validation scores are somewhat optimistic.

Table 21. Validation PR-AUC on four windows that precede the test seasons.

Validation window	Train years	Train rows	Val. races	LogReg PR-AUC	MLP PR-AUC
2014-2015	1950-2013	20,676	38	0.7164	0.7142
2016-2017	1950-2015	21,461	41	0.7282	0.7396
2018-2019	1950-2017	22,323	42	0.6893	0.7255
2020-2021 (used)	1950-2019	23,163	39	0.7301	0.7166

To measure the uncertainty directly we resampled whole validation races (Table 22). With 39 races, each model's PR-AUC has a 95% interval about 0.2 wide. The difference between the network and logistic regression is −0.013 with an interval from −0.040 to +0.015, and the forest's difference also includes zero. The data therefore do not show that any model beats the baseline, and we report the ranking as a tie within validation uncertainty. The chronological split stays because it is the only one with no same-race and no future leakage.

Table 22. Bootstrap over the 39 validation races. The last two rows are differences from logistic regression.

Comparison	Validation PR-AUC	95% low	95% high
Logistic regression	0.7301	0.6206	0.8378
Random forest	0.7125	0.5947	0.8299
Shallow neural network	0.7166	0.6045	0.8284
Random forest minus Logistic regression	−0.0176	−0.0553	0.0246
Shallow neural network minus Logistic regression	−0.0135	−0.0396	0.0147

We repeated the resampling on the unseen test seasons. It resamples predictions we already had and chooses nothing, and we ran it after every choice was frozen (Table 23). With 68 races each model's interval is about 0.11 wide. The forest minus logistic regression difference is +0.0150 (−0.012 to +0.046), and for the network it is +0.0134 (−0.016 to +0.044). Both intervals include zero, so the test seasons do not separate the models either, even though logistic regression scores 0.013 lower.

Table 23. Bootstrap over the 68 test races. The last two rows are differences from logistic regression.

Comparison	Test PR-AUC	95% low	95% high
Logistic regression	0.7023	0.6465	0.7606
Random forest	0.7173	0.6605	0.7780
Shallow neural network	0.7157	0.6619	0.7737
Random forest minus Logistic regression	0.0150	−0.0119	0.0462
Shallow neural network minus Logistic regression	0.0134	−0.0162	0.0444

8. Removing the grid hurts more than removing anything else

In our ablation we retrain logistic regression after deleting one feature group, then record the change in validation PR-AUC. We want to see which group the linear model was using, on the same rows as Section 7. Figure 11 shows the change from the full model. A bar to the left of zero is a loss.

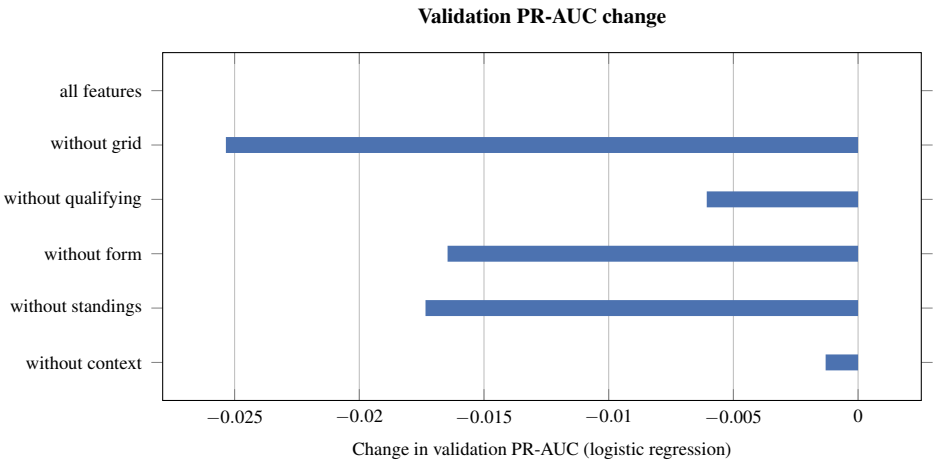


Figure 11. Change in validation PR-AUC when one feature group is removed from logistic regression. Zero is the full 16-feature model.

Table 24. Ablation of logistic regression on the 2020-2021 validation seasons.

Variant	Features kept	PR-AUC	ROC-AUC	PR-AUC change
all features	16	0.730	0.926	+0.000
without grid	12	0.705	0.920	−0.025
without qualifying	13	0.724	0.924	−0.006
without form	13	0.714	0.921	−0.016
without standings	12	0.713	0.924	−0.017
without context	14	0.729	0.925	−0.001

The full model sits at 0.730. Dropping the grid group costs 0.025 PR-AUC, the largest loss. Standings are next, at 0.017. Form, qualifying, and context cost 0.016, 0.006, and 0.001. The qualifying drop is small once the grid is already in the model. That fits the sport and the earlier definition: the published grid already contains qualifying order, plus any penalty. We do not conclude that qualifying is irrelevant on its own. We conclude that, next to the grid, it is mostly repeated information.

ROC-AUC barely notices the same deletion. It moves from 0.926 to 0.920 when the grid group is removed. This is the same lesson as the model comparison we drew in Section 7. A metric that stays high while the primary score drops is a poor guide for what to keep.

The context and qualifying changes are about the same size as the logistic-regression preprocessing gaps in Section 6. On 39 validation races, we treat those as small. The grid drop is several times larger, and it agrees with the descriptive ladder in Section 4.2, where podium rate falls from about 55% at the front of the grid to under 1% at the back.

> **Takeaway.** For the linear model, the final grid is the group we cannot remove without a visible loss. Recent standings help. Qualifying, form, and calendar context add less once the grid is present.

9. What the forest relies on, and where the two explanations differ

The explanations in this section are for the random forest, not for logistic regression, the model we selected in Section 7. Tree explanations apply directly to the forest. We do not borrow them as an account of the other models. Permutation importance shuffles one validation column at a time and measures the drop in PR-AUC. Global SHAP averages the size of each feature's push across 400 validation rows. Local SHAP is one of those rows. A positive value pushes that row toward a podium relative to the forest's background. A negative value pushes it away. These are associations inside the fitted forest. They are not estimates of what would happen if we moved a driver up the grid.

9.1 Permutation importance

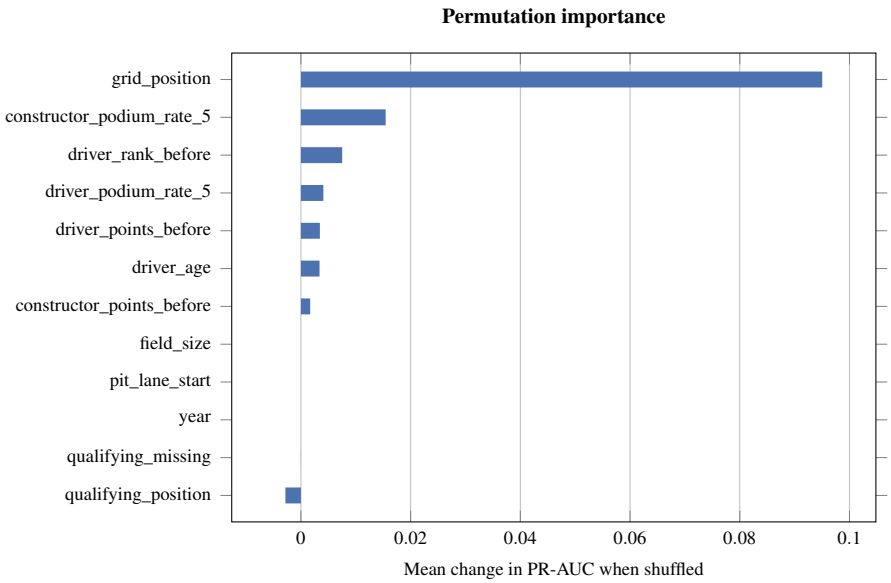


Figure 12. Permutation importance on a validation sample, scored with PR-AUC. A taller bar means shuffling that column hurt the forest more.

Table 25. Largest positive permutation importances, followed by the three most negative. Computed on the random forest and a validation sample.

Feature	Mean PR-AUC change when shuffled
grid_position	+0.095
constructor_podium_rate_5	+0.015
driver_rank_before	+0.008
driver_podium_rate_5	+0.004
driver_points_before	+0.003
driver_age	+0.003
constructor_points_before	+0.002
field_size	+0.000
constructor_rank_before	−0.009
front_row	−0.006
qualifying_time_seconds	−0.005

Grid position is in a class of its own, at +0.095. The next feature, constructor_podium_rate_5, is +0.015, about 16% of that, and every other feature is +0.008 or smaller. Driver podium rate is ranked 4 at +0.004, and year sits at +0.000. Shuffling constructor_rank_before changes PR-AUC by −0.009: the score rises slightly when the column is destroyed. We read a negative value as the forest not getting stable help from that column on these rows. We do not read it as evidence that the feature causes worse finishes.

9.2 Global and local SHAP

Figure 13 asks the same question with SHAP. Each validation prediction is split into one credit per feature, and we average the size of those credits. The sign is not in this chart. A tall bar means the forest moved the prediction a long way when that feature changed, whether the move was toward a podium or away from one.

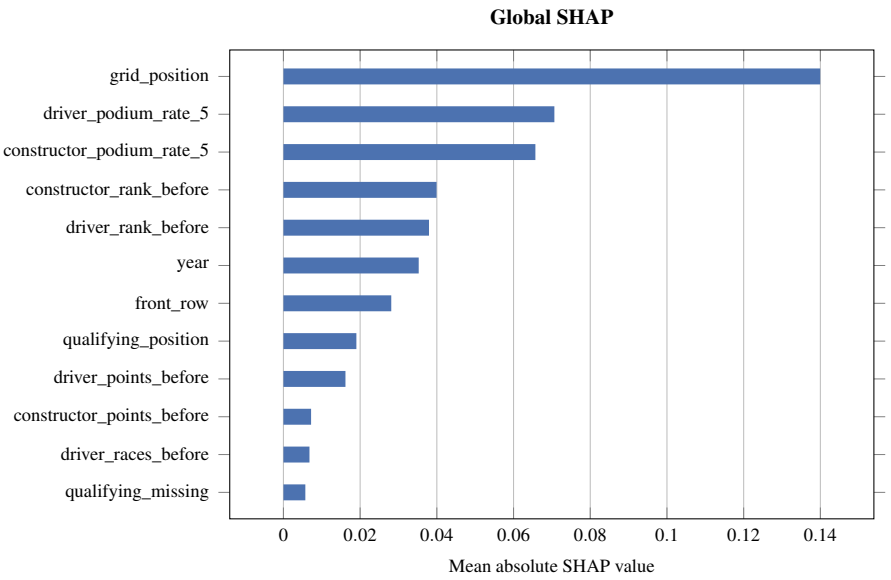


Figure 13. Mean absolute SHAP value by feature for the random forest, over 400 validation rows. This measures how large a feature's push is, not whether the push is positive.

Table 26. Eight largest mean absolute SHAP values for the random forest on the validation sample.

Feature	Mean absolute SHAP
grid_position	0.140
driver_podium_rate_5	0.071
constructor_podium_rate_5	0.066
constructor_rank_before	0.040
driver_rank_before	0.038
year	0.035
front_row	0.028
qualifying_position	0.019

Global SHAP also puts grid position first, at 0.140, and constructor podium rate second (0.066), the same top two as permutation importance. Driver podium rate follows at 0.071. The two methods differ below that. Year has a mean SHAP value of 0.035 (rank 6) but a permutation importance of +0.000, and driver points before the race is the reverse: rank 5 under permutation and rank 9 under SHAP. A reasonable reading is that year, form, and the standings overlap with each other and with the grid. The trees use them, but once the other columns are left intact, destroying one of them alone does not collapse the score. We treat the shared finding as the reliable one: grid position first, constructor form second.

Figure 14 is one validation row from that sample, not a row we picked because it looks tidy. It is George Russell, Williams, 2021 Dutch Grand Prix. He qualified and started 11th in a field of 20, sat 15th in the championship, and had a podium in one of his previous five starts (driver podium rate 0.2). He did not podium.

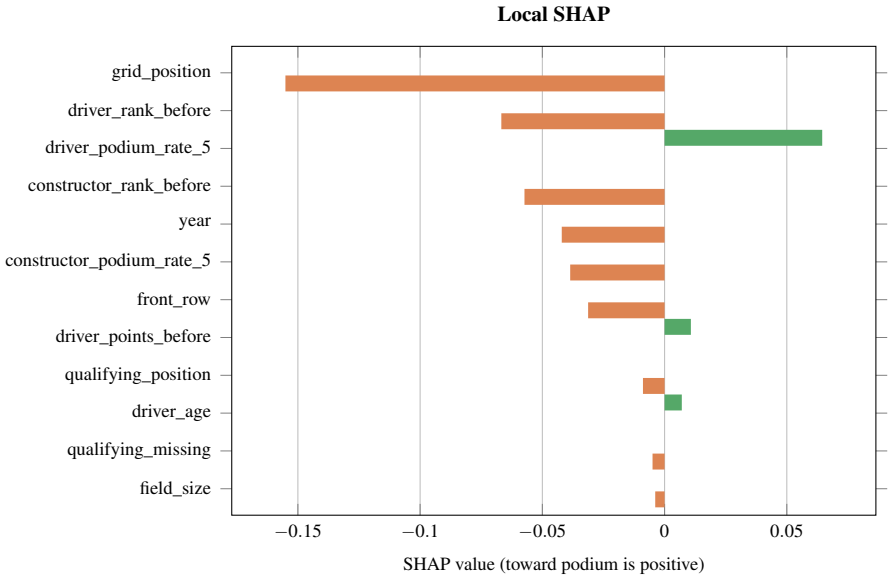


Figure 14. SHAP contributions for one validation row: Russell, Williams, 2021 Dutch Grand Prix, started 11th and did not podium. Negative values push the forest away from a podium.

Table 27. Largest SHAP contributions for the Russell validation row. The sign is the direction of the push.

Feature	SHAP value	Observed value
grid_position	−0.155	11.0
driver_rank_before	−0.067	15.0
driver_podium_rate_5	+0.064	0.2
constructor_rank_before	−0.057	8.0
year	−0.042	2021.0
constructor_podium_rate_5	−0.039	0.1

The largest push is grid_position = 11, and it is negative (−0.155). The next negative push is driver_rank_before = 15 (−0.067). The clearest positive push is driver_podium_rate_5 = 0.2 (+0.064), and it is smaller than the grid push. We observe a row where recent form points up and the start position points down, with the start position winning inside this forest. That is one example. It matches the ablation, and it matches the grid ladder in Section 4, but a single row is not a general law.

9.3 Local LIME explanation

LIME provides the required model-agnostic local explanation for the same fixed validation case. We create 5,000 nearby records by either retaining each observed feature or replacing it with a value drawn from the training seasons. The random forest scores those records, and a distance-weighted ridge model approximates the forest around the explained row. A positive coefficient pushes the local surrogate toward a podium; a negative coefficient pushes it away.

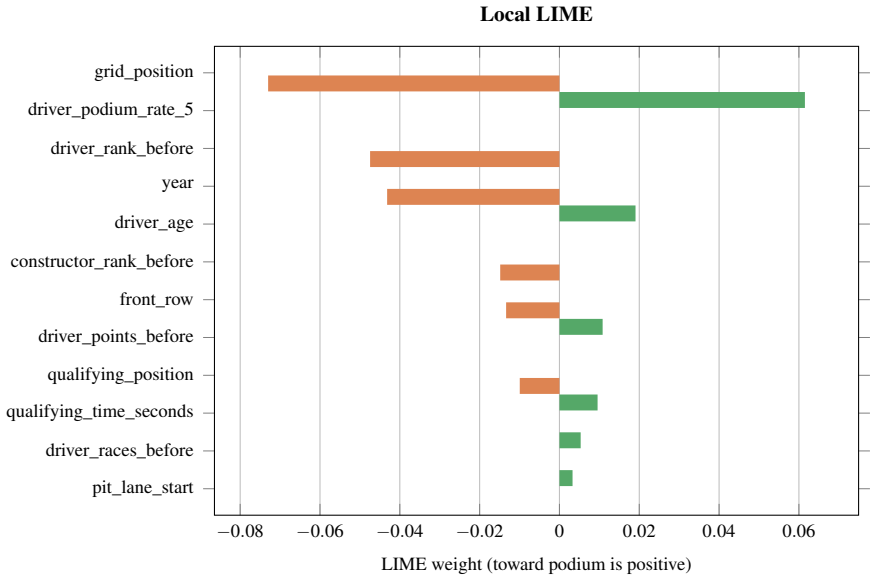


Figure 15. Local LIME contributions for one fixed validation prediction. The black-box probability is 0.174, the local surrogate estimate is 0.174, and weighted local R-squared is 0.211.

Table 28. Largest absolute LIME weights for the fixed validation row.

Feature	LIME weight	Observed value
grid_position	−0.073	11.0
driver_podium_rate_5	+0.062	0.2
driver_rank_before	−0.047	15.0
year	−0.043	2021.0
driver_age	+0.019	23.6
constructor_rank_before	−0.015	8.0
front_row	−0.013	0.0
driver_points_before	+0.011	13.0

The weighted local R-squared is 0.211. This fidelity value tells us how closely the simple surrogate follows the forest in the sampled neighborhood, so we read the local coefficients together with it rather than as global effects.

> **Takeaway.** Grid position is the feature the explanation methods agree on, with constructor form second under both. They differ on the smaller features, such as year and driver points, which is what we expect when form, standings, and year overlap with the grid. None of these values is a causal effect of starting further forward.

10. Two predictions from the selected model

The examples use the selected model, logistic regression, because it had the best validation PR-AUC. Its frozen cutoff is 0.625. Missing inputs are filled by the imputer that was fit on the training years. We show one complete historical row and one partial record that was never a real entry.

The complete row is Lewis Hamilton, Mercedes, 2022 Bahrain Grand Prix, round 1. He qualified 5th and started 5th in a field of 20. His previous five starts in this table were all podiums (driver rate 1.0) and the constructor's recent podium rate was 0.7. Because it is the first race of the season, his and Mercedes' championship points and rank are both 0. Logistic regression gives this row a podium probability of 0.906. That is above 0.625, so the predicted label is a podium. The actual result was third, a podium, so the prediction is correct. A fifth-place start and a perfect recent podium run were enough to lift the probability over the cutoff. The row is picked automatically as the first complete 2022 row. Before the round-1 standings fix, round-1 rows had empty standings and were skipped, so the example used to be the 2022 Saudi Arabian Grand Prix.

The partial record supplies only what we chose to type in: grid position 3, not a pit-lane start, not a front-row start, a field of 20, and season 2024. Every other feature is missing and is replaced with the training median. Logistic regression returns 0.387, below the cutoff, so the predicted label is not a podium. This record has no true result, because it is not a driver who started a race.

We should not read 0.387 as the historical podium rate of cars that start third. Section 4.2 puts that rate near 55% for grid positions 1-3. The partial row is a third-place start glued onto a median historical driver, and the median driver in a 1950-2019

training set is not a podium contender. The low probability is what this imputer does with a half-empty row. It is evidence that the function accepts missing fields. It is not evidence about the front of the grid.

> **Takeaway.** On a real 2022 row, the selected model correctly predicted a podium for a fifth-place start backed by a perfect recent podium run. On a synthetic partial row, the number is an imputed profile, and we do not interpret it as a base rate.

11. What we do not want the results to be taken to say

Qualifying coverage is a property of this snapshot. We described it, and we kept a missingness flag, but we cannot tell from the chart alone how much of the pre-1994 gap is history and how much is the way the public tables were filled.

Constructor form is built per constructor identifier, and a renamed team usually gets a new one, so its form starts empty: Toro Rosso became AlphaTauri in 2020, Renault became Alpine in 2021, and AlphaTauri became RB in 2024. A few old identifiers are also reused years later, so Aston Martin in 2021 is linked to the unrelated 1959-1960 Aston Martin. Standings restart every season and driver form follows the driver, so only constructor form is touched, and only in the first five races of such a season: 30 of 780 validation rows and 20 of 1,359 test rows. None of them is a podium, and their constructor form is 0 or empty, which the imputer fills with the training median of 0. We keep the identifiers as they are rather than hand-map renames, because each link is a judgement (a rebrand is not the same as a sale to a new owner), and the evaluation would barely move.

The home analysis depends on a nationality-to-country map. Drivers we could not map are excluded, and drivers with more than one sporting nationality are forced into one label. The small home samples in the front of the grid mean a one-point gap is not something we would defend as a precise estimate.

The mechanical list is a rule we wrote down, not a measurement from the car. Ambiguous statuses can fall outside it. Leading rates such as Caterham's rest on a few dozen starts. The lower 2022-2024 totals are consistent with a more reliable field, and they are also consistent with changes in how retirements were described.

Validation is 39 races. We used it to pick a cutoff and a model, which is what a held-out test set requires, and we do not pretend it can rank effects of a few thousandths. A bootstrap over those races gives each PR-AUC an interval about 0.2 wide, and the differences between the models fall inside it. On the unseen test seasons PR-AUC is 0.702 for logistic regression, 0.717 for the forest and 0.716 for the network. Logistic regression is 0.013 behind, but a bootstrap over the 68 test races puts both differences from it inside an interval that includes zero (Section 7.2). We therefore do not claim that the more complex models are better. We treat the three as tied in ranking quality and keep logistic regression as the readable baseline. Regulation changes in 2014 and 2022 sit inside the eras we compared. The models are associating historical patterns. They do not tell a team what would happen if it started one place further forward, and a probability is not a guarantee about the next race.

12. What the evidence supports

The tables join, and after 85 shared-drive groups are resolved and 1,366 did-not-enter rows are removed we have one row per driver per race, 25,302 of them, with a podium rate of 13.4%. Qualifying is missing for most of the early history and nearly complete from 2003, so the grid is the starting-order feature we can actually follow from 1950.

Descriptively, front-row cars podium far more often than the field, and more often again after 2013. Home nationality, inside a grid band, does not produce a consistent extra podium rate. Mechanical retirement rates, on our component list and among teams with at least 20 starts, are lower in 2022-2024 than in 2014-2021, and the highest rates belong to shorter entries.

Predictively, logistic regression, a random forest, and a shallow network all clear the PR-AUC baseline of the recent seasons by a wide margin. Logistic regression is the one we select from the validation scores, by seed 42 and by the three-seed average. The untouched test seasons put the forest and the network slightly ahead (0.717 and 0.716, against 0.702), but the three models are within bootstrap uncertainty of each other on both validation and test, so the simplest and most readable model is the one we keep. Ablation, permutation, and SHAP agree that grid position carries the prediction, and permutation and SHAP also agree that constructor form comes second. They differ on the smaller features, and we do not force them to agree. The Hamilton example fits this picture in one row: a fifth-place start backed by a recent run of podiums lifted him over the cutoff, and he finished third.